

ELEC 475: Computer Vision

Lab 4

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1 Introduction

Contrastive Language-Image Pre-training (CLIP) enables flexible zero-shot classification, retrieval, and image-text reasoning through a joint embedding space that describes the relationships between images and text captions. This document details the construction and training of an image encoder used alongside a pretrained text transformer, forming a CLIP-style model.

2 Methodology

The following section details the model architecture, dataset preparation, and training.

2.1 Vision Model Design

The vision model consists of a ResNet50 image encoder followed by a projection head that maps image features into a 512-dimensional embedding space. The ResNet50 encoder was initialized with pretrained ImageNet weights, and the final classification layer was removed. Instead, the output of the ResNet encoder was connected to a projection head, consisting of two linear layers with Gaussian Error Linear Unit (GELU) activation functions. The output of the projection head produces the embedded output for the input image.

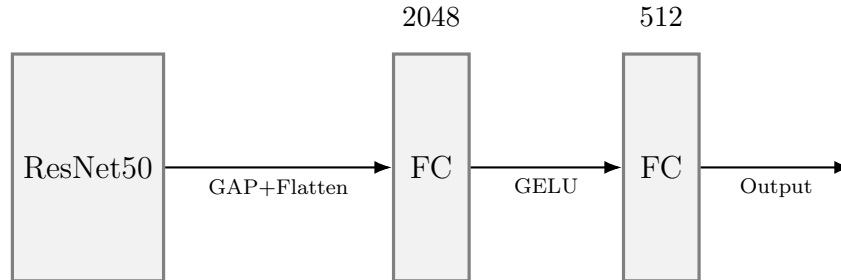


Figure 1: Vision Model Diagram

2.2 Dataset & Preprocessing

Model training and validation were performed on the MS-COCO 2014 dataset, which includes approximately 80,000 training images and 40,000 validation images. All 80,000 training and 40,000 validation images were used during the training of the Image Encoder. Before being passed to the model, each image was resized to 224×224 and normalized using the CLIP mean and standard deviation.

Each image in the COCO 2014 dataset is associated with five descriptive captions. Before training, text captions were preprocessed using the CLIP-VIT-base-patch32 model, provided by OpenAI. This model provided both the tokenizer and text transformer required to encode the image captions into the 512-dimensional feature space. Pre-encoding image captions is not necessary, but was performed to improve the training speed of the image model. Using pre-encoded captions is only possible when the text transformer is kept frozen during model training, as is the case in this lab.

Note:

2.3 Image Encoder Training

The image encoder, explained in Section 2.1, was trained over 80 epochs with a batch size of 150. The projection head was trained with a learning rate of 0.008. To avoid knowledge loss, the ResNet50 backbone was trained with a learning rate of 0.0007. Both networks used a learning decay of 0.0003, and the total loss was calculated using InfoNCE loss with a temperature of 0.04.

The loss function, InfoNCE loss, is a contrastive loss function used to bring matching pairs of data closer in an embedding space while pushing non-matching pairs further apart. In CLIP-style training, a positive pair is an image and its corresponding caption. Negative pairs occur between images and all other captions not paired with that image. This is computed by comparing the cosine similarity between different points. A low loss value occurs when the cosine similarity is high for related points and low for unrelated points. In InfoNCE loss, the temperature value (τ) controls the sharpness of the cosine similarity. Lower values of τ generate sharper similarity values, aggressively forcing the model to separate unrelated embeddings.

Rather than training the model on all captions for every image in each training step, a single caption was randomly selected from the set of five captions associated with each image. Selecting one caption reduces the computational cost of training, while random selection ensures that the model is exposed to all captions at some point during the training. Additionally, random selection can help prevent overfitting an image to one of its captions.

2.4 Augmentation & Modification

To improve model accuracy, two modifications were made to the training process. These modifications were evaluated individually and in combination through an ablation study.

2.4.1 Training Image Augmentation

The first modification involved data augmentation applied to the training images to increase dataset variability and prevent overfitting. Each image was randomly cropped and horizontally flipped. Additional augmentations included color jitter, which adjusts the brightness, contrast, saturation, and hue, and Gaussian blur to slightly alter image sharpness. All augmentations were applied before resizing and standardizing the image according to the CLIP values. These transformations expose the model to varied visual conditions, encouraging the encoder to learn more robust and generalizable features.

2.4.2 Temperature Scheduling

The second modification involved implementing a temperature scalar in the InfoNCE loss. As explained in Section 2.3, the temperature parameter τ controls the sharpness of the contrastive distribution by scaling the similarity scores between image and text embeddings. Adjusting this value during training (with a scheduler) allows the model to initially tolerate more ambiguity in the embeddings and gradually enforce stricter separation between positive and negative pairs. This can improve convergence and stabilize training by reducing large fluctuations in loss early in training.

3 Results

In total, four models were trained and evaluated: the base model without any modifications, the model with data augmentation, the model with temperature scaling, and the model with a combination of both the data augmentation and temperature scaling modifications.

Additionally, the models were assigned to visualize the top-5 retrievals for the 'sport' query, and return the scores for each image. Finally, the models were asked to analyze the image shown in Figure 2, which contains a dog and shoes within a basket. Each model was tasked with returning the scores for the classifications: "a dog," "a shoe," and "a basket."



Figure 2: Test Image - COCO 2014 Image # 42

3.1 Base Model

During training, the base model (without modifications) developed the loss curve shown in Figure 3.

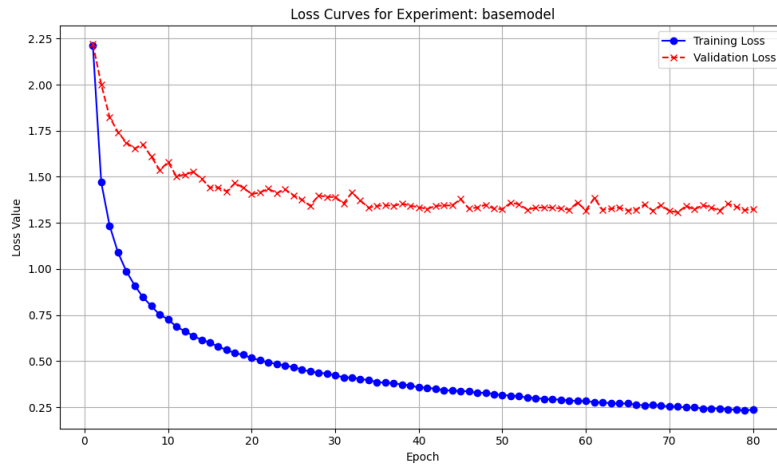


Figure 3: Base Model Loss Curve

Following training, the model was evaluated using the Recall@K metric and validation dataset. Results are shown in Table 1.

Table 1: Base Model Recall Values

Recall@ k	Image to Text	Text to Image
Recall@1	5.77%	4.86%
Recall@5	16.05%	13.96%
Recall@10	23.12%	20.55%

Using the query "sport", the top five returned results are shown in Figure 4.

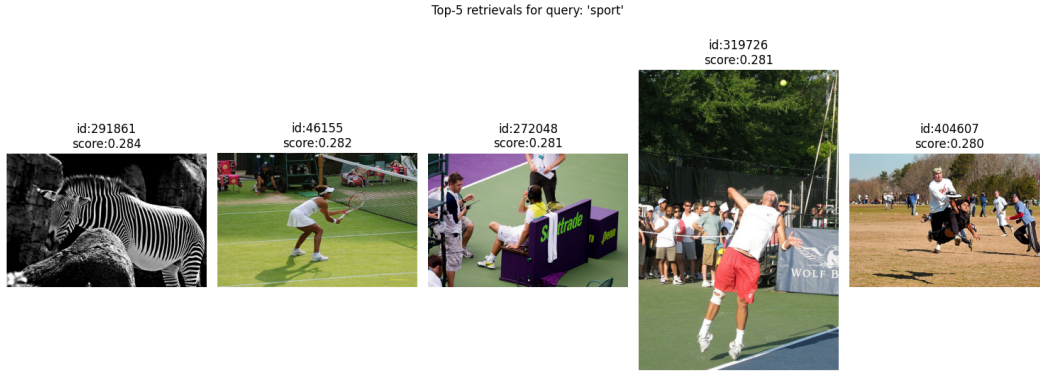


Figure 4: Base Model 'sport' query results

Finally, the model was evaluated using the test image shown in Figure 2. The resultant scores produced by the model are shown in Table 2.

Table 2: Base Model Sample Classification Image Evaluation

Query	Predictive Score
"a dog"	0.1403
"a shoe"	0.1524
"a basket"	0.3164

3.2 Data Augmentation

During training, the data augmentation model developed the loss curve shown in Figure 5.

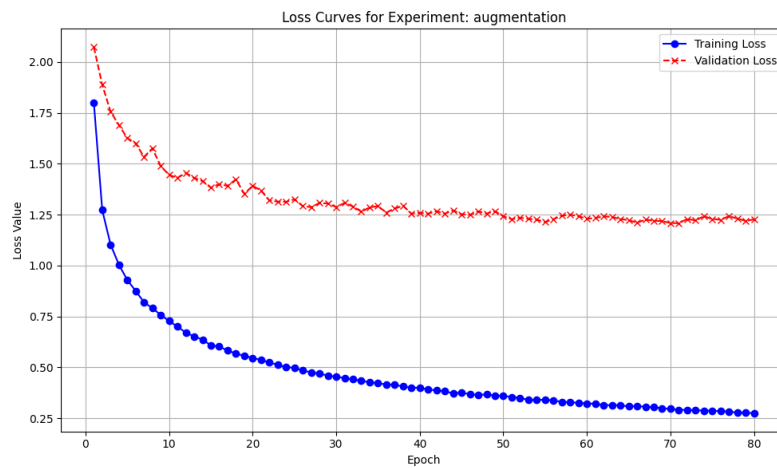


Figure 5: Data Augmentation Model Training Loss Curve

Following training, the model was evaluated using the Recall@K metric and validation dataset. Results are shown in Table 3.

Table 3: Data Augmentation Model Recall Values

Recall@k	Image to Text	Text to Image
Recall@1	7.44%	6.36%
Recall@5	20.03%	17.66%
Recall@10	28.75%	25.52%

Using the query "sport", the top five returned results are shown in Figure 8.



Figure 6: Data Augmentation Model 'sport' query results

Finally, the model was evaluated using the test image shown in Figure 2. The resultant scores produced by the model are shown in Table 6.

Table 4: Data Augmentation Model Sample Classification Image Evaluation

Query	Predictive Score
"a dog"	0.1264
"a shoe"	0.2696
"a basket"	0.3048

3.3 Temperature Scheduling

During training, the temperature scaling model developed the loss curve shown in Figure 7.

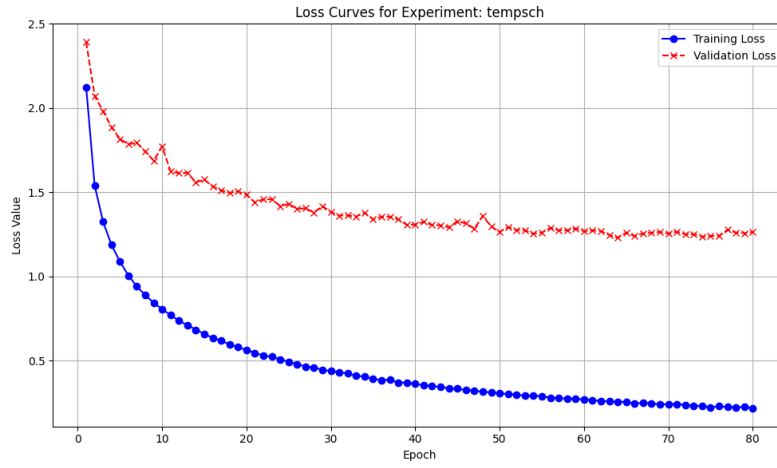


Figure 7: Temperature Scaling Model Training Loss Curve

Following training, the model was evaluated using the Recall@K metric and validation dataset. Results are shown in Table 5.

Table 5: Temperature Scaling Model Recall Values

Recall@ k	Image to Text	Text to Image
Recall@1	6.23%	5.18%
Recall@5	17.14%	14.97%
Recall@10	24.68%	21.83%

Using the query "sport", the top five returned results are shown in Figure 8.



Figure 8: Temperature Scaling Model 'sport' query results

Finally, the model was evaluated using the test image shown in Figure 2. The resultant scores produced by the model are shown in Table 6.

Table 6: Data Augmentation Model Sample Classification Image Evaluation

Query	Predictive Score
"a dog"	0.1104
"a shoe"	0.2974
"a basket"	0.2674

3.4 Combined Modifications

The combined modifications model, which combines data augmentation with temperature scaling, produced the loss curve shown in 9.

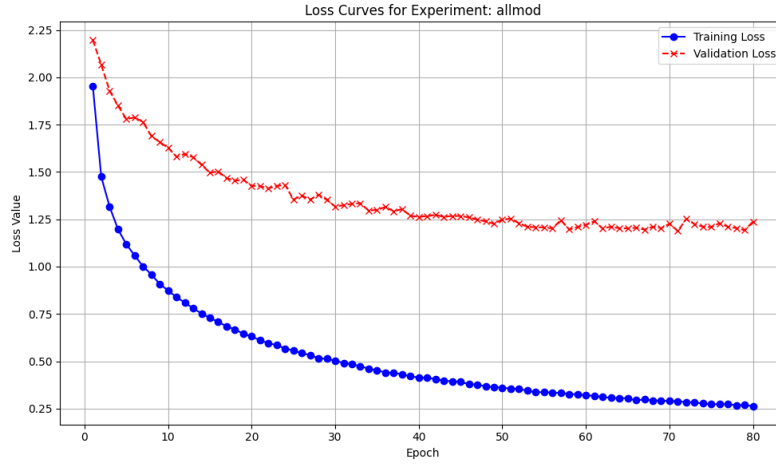


Figure 9: Combined Model Training Loss Curve

Following training, the model was evaluated using the Recall@K metric and validation dataset. Results are shown in Table 7.

Table 7: Combined Model Recall Values

Recall@ k	Image to Text	Text to Image
Recall@1	7.76%	6.37%
Recall@5	20.56%	17.58%
Recall@10	29.11%	25.44%

Using the query "sport", the top five returned results are shown in Figure 10.

Finally, the model was evaluated using the test image shown in Figure 2. The resultant scores produced by the model are shown in Table 8.



Figure 10: Temperature Scaling Model 'sport' query results

Table 8: Combined Model Sample Classification Image Evaluation

Query	Predictive Score
"a dog"	0.1442
"a shoe"	0.2536
"a basket"	0.3341

4 Discussion & Analysis

4.1 Ablation Study

The evaluation of each model included a Recall@K derivation for both image-to-text and text-to-image retrieval. See Table 9 for the ablation study.

Table 9: Ablation Study

Model	K	Metric	Recall (%)	Min	Max	Mean	Std
Base	1	I2T@1	5.77	0.00	100.00	5.77	23.33
	1	T2I@1	4.86	0.00	100.00	4.86	21.50
	5	I2T@5	16.05	0.00	100.00	16.05	36.70
	5	T2I@5	13.96	0.00	100.00	13.96	34.66
	10	I2T@10	23.12	0.00	100.00	23.12	42.16
	10	T2I@10	20.55	0.00	100.00	20.55	40.41
Data	1	I2T@1	7.44	0.00	100.00	7.44	26.25
	1	T2I@1	6.36	0.00	100.00	6.36	24.40
	5	I2T@5	20.03	0.00	100.00	20.03	40.02
	5	T2I@5	17.66	0.00	100.00	17.66	38.14
	10	I2T@10	28.75	0.00	100.00	28.75	45.26
	10	T2I@10	25.52	0.00	100.00	25.52	43.60
Temp	1	I2T@1	6.23	0.00	100.00	6.23	24.17
	1	T2I@1	5.18	0.00	100.00	5.18	22.17
	5	I2T@5	17.14	0.00	100.00	17.14	37.68
	5	T2I@5	14.97	0.00	100.00	14.97	35.67
	10	I2T@10	24.68	0.00	100.00	24.68	43.12
	10	T2I@10	21.83	0.00	100.00	21.83	41.31
Combined	1	I2T@1	7.76	0.00	100.00	7.76	26.76
	1	T2I@1	6.37	0.00	100.00	6.37	24.41
	5	I2T@5	20.56	0.00	100.00	20.56	40.41
	5	T2I@5	17.58	0.00	100.00	17.58	38.06
	10	I2T@10	29.11	0.00	100.00	29.11	45.43
	10	T2I@10	25.44	0.00	100.00	25.44	43.55

4.2 CLIP Loss

The Contrastive Language-Image Pre-training (CLIP) objective is to align the image and text embedding spaces by teaching the model to simultaneously associate matching image-caption pairs and dissociate matching pairs from all other pairs. This is achieved through a symmetric contrastive loss that operates over anchors, positives, and negatives.

In each batch, images and their corresponding captions for the set of *positive pairs*. For any given image embedding v_i , its paired caption embedding t_i serves as the unique positive sample. All remaining captions $\{t_j\}_{j \neq i}$ act as *negatives*, providing contrastive examples that the loss function uses to push dissimilar items apart in the representation space. The same structure applies in the opposite direction when treating each caption as the anchor and all images as candidate matches.

Geometrically, the embeddings are normalized to lie on a hypersphere, making the cosine similarity directly represent angular closeness. The loss, therefore, encourages paired embeddings to move towards each other by reducing the angle between them, while non-matching pairs are pushed further apart. Because every batch provides many negatives, the model benefits significantly from large batch sizes, which supply a richer set of contrasts and more effectively shape the embedding space.

4.3 Training Outcome

The training was done on a machine with a NVIDIA RTX 3080 and an AMD Ryzen 7 3700X. Each model took 3.5-4 hours to train fully. The hyperparameters used for each model are shown in Table 10:

Table 10: Model Configuration Parameters Across Different Scenarios

Parameter	Base Config	Data Augmentation	Temp. Scaling	Combined
epochs	80	80	80	80
backbone_lr	0.0007	0.0005	0.0006	0.0005
head_lr	0.008	0.005	0.006	0.005
weight_decay	0.0003	0.0003	0.0003	0.0003
batch_size	150	128	150	150
projection_dim	512	512	512	512
temperature	0.04	0.04	0.07	0.07
temp_decay	—	—	0.985	0.985
freeze_backbone	false	false	false	false
num_workers	16	16	16	16

The training losses across all four models showed significantly higher validation loss over training loss. Regarding the best validation losses from training, the strongest model proved to be the data augmentation model for final validation loss, and the combined modifications model for best validation

loss.

4.4 Recall Evaluation

Recall@K is an evaluation metric used to measure how well a model retrieves the correct item when searching within a larger set. In retrieval tasks such as image-text matching, the model produces a list of ranked choices based on similarity. The k parameter determines how many results the model will return, and Recall@K quantifies how often the true match appears within the top k positions of that ranking.

The Base model achieves the lowest overall recall across all metrics, with I2T@1 at 5.77% and T2I@1 at 4.86%. Increasing K improves recall as expected, but it remains the lowest among all ablation variants, showing that neither additional data nor temperature scaling is applied.

The Data model demonstrates a clear improvement over Base, particularly in I2T@1 (7.44%) and T2I@1 (6.36%), reflecting the positive effect of the data augmentation/modification. Gains are also consistent across higher Ks, with I2T@10 at 28.75% and T2I@10 at 25.52%, indicating stronger retrieval performance overall.

The Temp model shows moderate improvements over Base, with I2T@1 at 6.23% and T2I@1 at 5.18%, suggesting that temperature scaling alone provides some benefit but less than data augmentation. Recall gains at higher Ks are smaller than the Data model, with I2T@10 at 24.68% and T2I@10 at 21.83%.

The Combined model, which integrates both data modifications and temperature scaling, consistently achieves the highest recall across almost all metrics. I2T@1 reaches 7.76% and T2I@1 6.37%, with the highest recalls at K=5 and K=10 (I2T@10 29.11%, T2I@10 25.44%). This indicates that combining the two modifications yields an additive effect, outperforming either change in isolation.

4.5 Semantic Alignment between Modalities

The results for each model were very reasonable for their alignment. For the top-5 retrievals of the 'sport' query, the temperature scaling model held the best scores, followed by the combined modifications model, the base model, and lastly the data augmentation model. Even though the data augmentation model performed the worst in scoring, it can be said that the base model provided poorer results, with the strongest score assigned to an image that does not depict a sport even remotely. Additionally, it can further be said that the inclusion of temperature scaling in training boosted the classification of a query, while data augmentation poorly affected it, as the temperature scaling model performed the strongest, and not the combined modifications model.

In regards to the classification of an image, the results are less straight-forward. The model with the strongest score for labeling "a dog" is the combined modifications model, for labeling "a shoe" is the temperature scaling model, and for labeling "a basket" is the combined modifications model. As the combined modifications model strengthened the scores of the base model, the results of the data augmentation model and the temperature scaling model show that the independent modifications affected the scoring separately, and only their combination yielded the strongest results.

5 Conclusion

In retrospect, the success of this lab is conflictive. All four model variations yielded poor Recall@K values, way below even the 50% line. Additionally, the loss curves showed clear signs of overfitting, as training loss in all four models is considerably lower than the validation loss. However, the retrievals for the 'sport' query yielded strong results, as the top 5 images are strong for each model, with only two of the four models (base and data augmentation models) having one incorrect classification. Nevertheless, the classification of the image also yielded poor results. Overall, although the results of the model are poor, the performance of the image retrieval reveals that poor numerical scores do not always indicating a failing model.

Some modifications that could be done to the model includes the use of a stronger encoder architecture, such as the ViT-B/32, or an EfficientNet-style backbone for better features. Additionally, the addition of a trainable text encoder would allow for simultaneous end-to-end training, and would yield tighter alignment. Finally, developing the projection head to be more robust (layer-norm, residuals, dropout, etc.) would also improve the performance.